

ANN → structured data
semi-structured data
delimited files } → Supervised learning problems

CNN → Dealing with image dataset
Image Classification
Transfer Learning

Deal with Sequence data } → not just about data, its data + order (sequence)

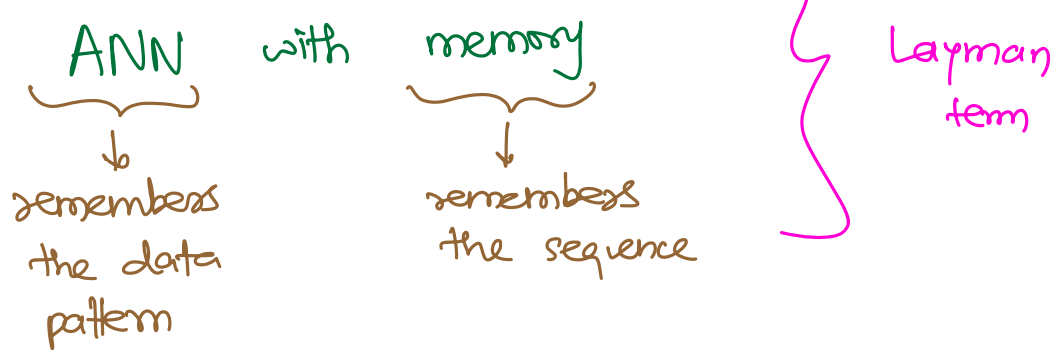
① Time-series data
- Sales forecasting } → 20000
- Share price prediction 2010

② Language Translation / Machine Translation } → my name is PN
HAI HAI PN

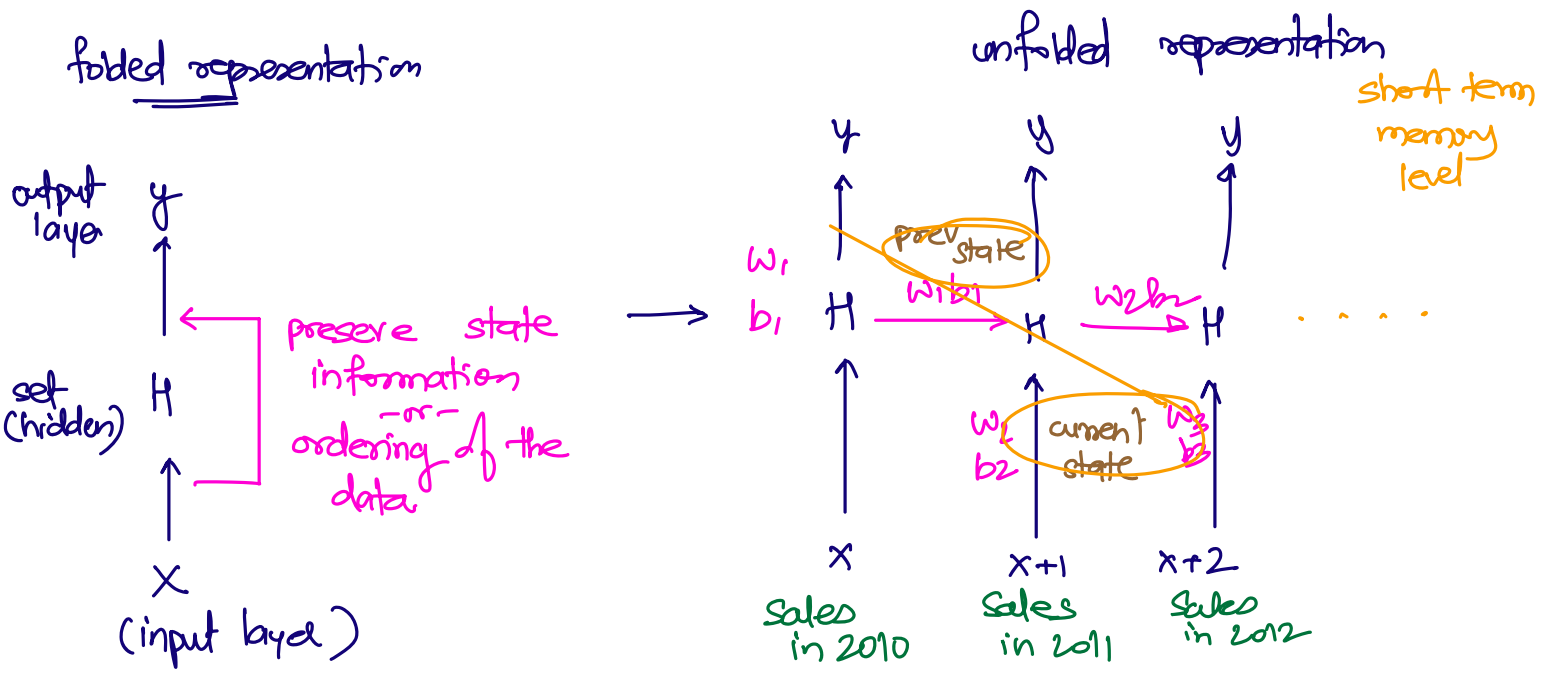
③ Image Captioning
my name PN is

 → model → a boy smiling

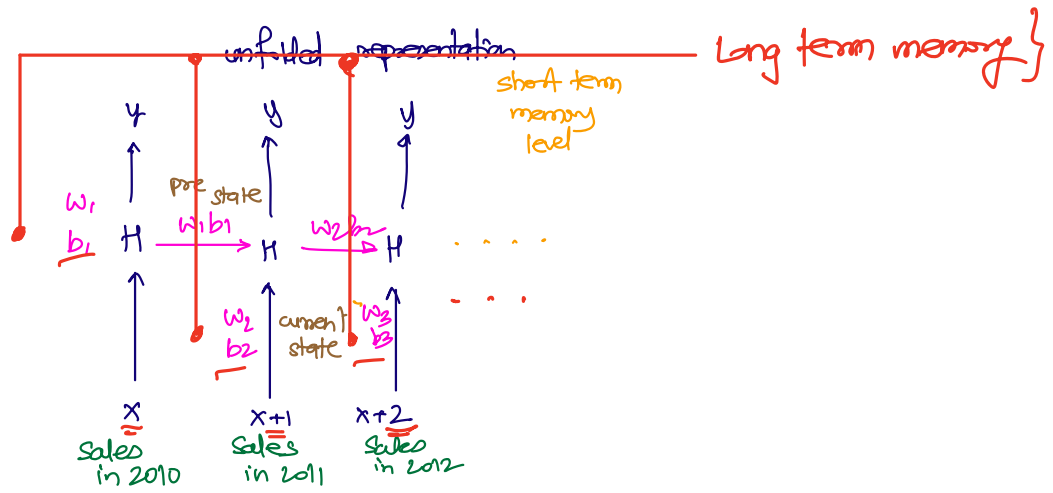
RNN (Recurrent Neural Network)



RNN has a mechanism of preserving the sequence of the data.



LSTM (Long Short Term Memory)



Transformer

- short term
- long term
- position of each entity (position embedding)
- attention mechanism

eg:

sales = [0.1, 0.15, 0.2, 0.25, 0.3, 0.35] } scaling

seqLength = 3 (how many states to consider to predict the next state)

next state prediction

↓
next token prediction

X	Y
0.1, 0.15, 0.2	0.25
0.15, 0.2, 0.25	0.3
0.2, 0.25, 0.3	0.35
0.25, 0.3, 0.35	null

Generative Models

Auto Encoders

Data Regeneration | Data Compression | Data Transmission

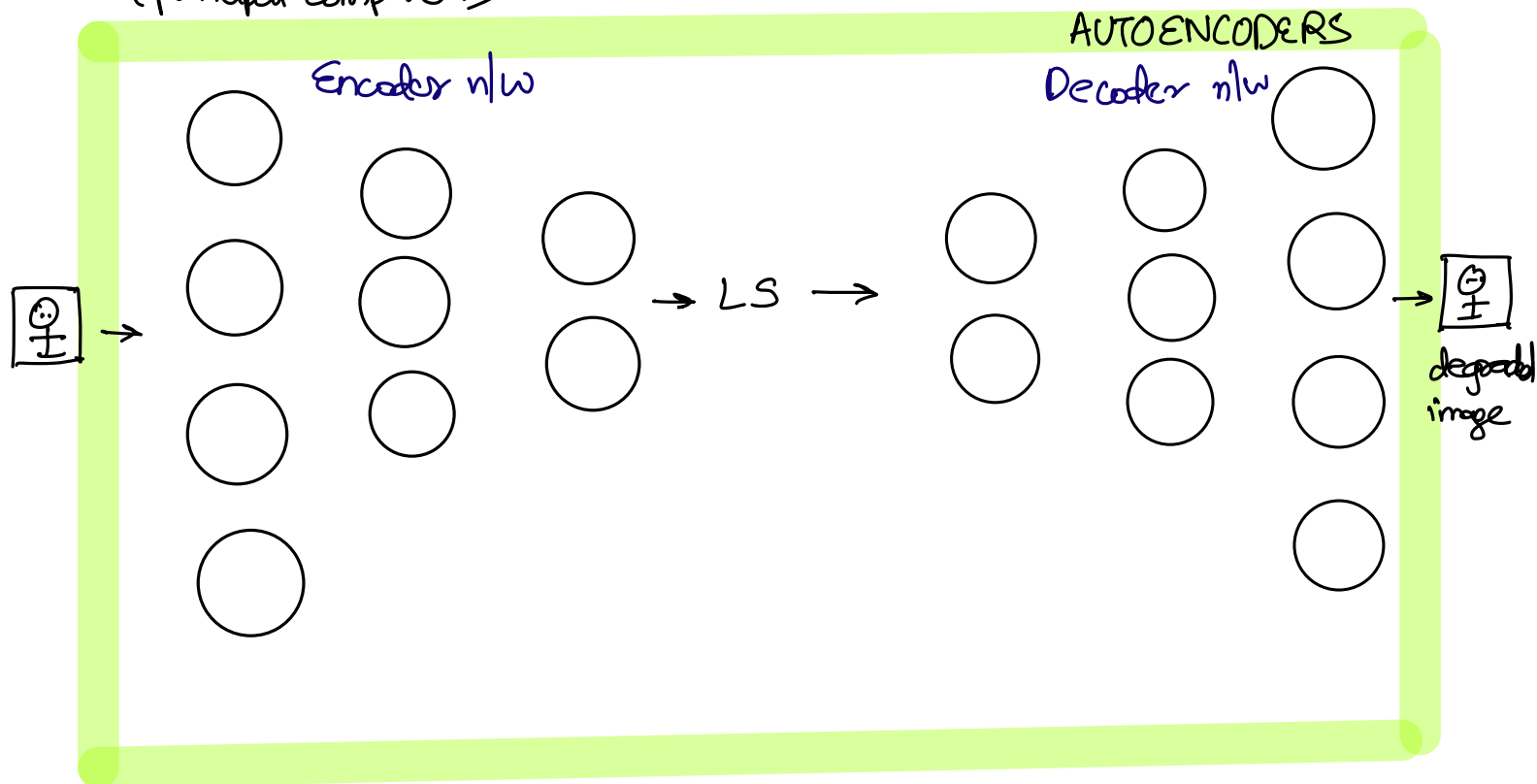
Autoencoders comprises of two networks

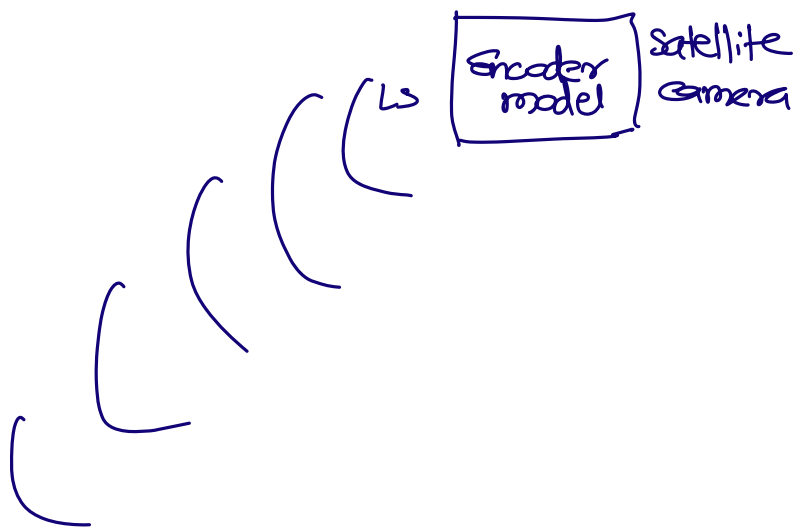
Encoder
network

Goal: accept i/p data and
convert the same into
LATENT SPACE (LS)
(principal component)

Decoder
network

accept the Latent Space
and regenerate the i/p.





NASA ~~A~~

